

MATH 3330 Homework 5

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Question 1

Let X be an ordered set in the order topology. Show that $\overline{(a, b)} \subseteq [a, b]$. Under what conditions does the equality hold?

Proof: Let X be an ordered set in the order topology. Let $x \in \overline{(a, b)}$, so every open neighborhood of x must intersect (a, b) . Then consider the following:

If $x < a$, then we can choose some m, n such that $m < x < n < a$. Then, $x \in \overline{(m, n)}$, but $(m, n) \cap (a, b) = \emptyset$ since $m < n < a$, contradicting the fact that (m, n) must intersect (a, b) if $x \in \overline{(a, b)}$. Thus $x \not< a$, so $a \leq x$.

Similarly, if $b < x$, then we can choose some m, n such that $b < m < x < n$. Then, $x \in \overline{(m, n)}$, but $(m, n) \cap (a, b) = \emptyset$ since $b < m < n$, contradicting the fact that (m, n) must intersect (a, b) if $x \in \overline{(a, b)}$. Thus $b \not< x$, so $x \leq b$.

So, $a \leq x \leq b$, or equivalently, $x \in [a, b]$. Therefore, $\overline{(a, b)} \subseteq [a, b]$.

Under what conditions does the equality hold?

Equality will hold when $[a, b] \subseteq \overline{(a, b)}$. This means that both $a, b \in \overline{(a, b)}$, so every neighborhood of a and every neighborhood of b contains a point in (a, b) . In the order topology, neighborhoods of a contain intervals of the form (u, v) with $u < a < v$. Thus $a \in \overline{(a, b)}$ exactly when every interval contains a point x with $a < x < b$. This holds precisely when there is no element of X that is the immediate successor of a . Otherwise, if c were the smallest element greater than a , we could choose an interval around a that lies entirely below c and it would not intersect (a, b) .

Similarly, $b \in \overline{(a, b)}$ exactly when there is no element of X that is the immediate predecessor of b .

Therefore, $\overline{(a, b)} = [a, b]$ when a has no immediate successor and b has no immediate predecessor in X .

Question 2

Let A_α be a subset of a space X . Decide if $\overline{\cap A_\alpha} = \cap \overline{A_\alpha}$. If not, is one a subset of the other?

The equivalency is not true, and $\overline{\cap A_\alpha} \subseteq \cap \overline{A_\alpha}$.

Proof: Let $x \in \overline{\cap A_\alpha}$. Then every open neighborhood of x intersects with $\cap A_\alpha$. Then, each of those neighborhoods must also intersect all individual A_α since those neighborhoods intersect with the intersection of each A_α . So, $x \in \overline{A_\alpha}$ for all α , and then $x \in \cap \overline{A_\alpha}$. Therefore, $\overline{\cap A_\alpha} \subseteq \cap \overline{A_\alpha}$.

Now for $\cap \overline{A_\alpha} \subseteq \overline{\cap A_\alpha}$, consider the counterexample for when $A_1 = (0, 1)$ and $A_2 = (1, 2)$. Then, $\overline{(0, 1)} \cap \overline{(1, 2)} = \{1\}$ and $\overline{(0, 1) \cap (1, 2)} = \emptyset$. Since $\{1\} \not\subseteq \emptyset$, $\cap \overline{A_\alpha} \not\subseteq \overline{\cap A_\alpha}$.

Therefore, $\overline{\cap A_\alpha} \neq \cap \overline{A_\alpha}$, however $\overline{\cap A_\alpha} \subseteq \cap \overline{A_\alpha}$.

Question 3

In the finite complement topology on $\mathbb{R}$, to what point or points does the sequence $x_n = \frac{1}{n}$ converge?

A sequence x_n converges to x if for every open set $U \subseteq \mathbb{R}$ containing x , all but finitely many x_n are in U . Let $U \subseteq \mathbb{R}$ be an open set containing $x \in \mathbb{R}$. In the complement topology U is open if $U = \emptyset$ or $U = X - F$ where F is a finite set in $\mathbb{R}$. $U \neq \emptyset$ since U contains x , so then $x \in X$, but $x \notin F$.

Since F is finite and the sequence x_n takes in distinct values that are each assigned to unique values in $\{\frac{1}{n}\}$, there are only finitely many indices that $x_n \in F$. Therefore, there must exist some N such that for all $n > N$, $x_n \notin F$, and since $U = X - F$, $x_n \in U$. Then, all but finitely many terms of x_n converge to x . Since x was arbitrary, the sequence converges to every $x \in \mathbb{R}$.

Question 4

Show that the T_1 axiom is equivalent to the condition for each pair of points of X , each has a neighborhood not containing the other.

The T_1 axiom states that a space X is T_1 if for each pair $x \neq y$, there exists an open set containing x but not y , and an open set containing y but not x . Additionally, a neighborhood of x is a set U that contains an open set containing x .

Proof: Let X be T_1 and $x, y \in X$ such that $x \neq y$. By the definition of a T_1 space, there must exist an open set U containing x that does not contain y , as well as an open set V containing y that does not contain x . Thus U is a neighborhood of x that doesn't contain y , and V is a neighborhood of y that doesn't contain x . Therefore, each pair of points x and y have some neighborhood not containing the other.

Now assume that each pair of non-equal points x and y have a neighborhood not containing the other. Then by the definition of a neighborhood, $x \in U \subseteq X$ and $y \in V \subseteq Y$. Therefore, there is an open set containing x but not y and another open set containing y but not x , so X is T_1 .

Question 5

Prove that for functions $f : \mathbb{R} \rightarrow \mathbb{R}$, the $\epsilon - \delta$ definition of continuity implies the open set definition.

Proof: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. f is continuous by the $\epsilon - \delta$ definition of continuity at a point x if for every $\epsilon > 0$, there exists some $\delta > 0$ such that given another point y that's close to x , $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$. We must show that given the $\epsilon - \delta$ definition of continuity, f is continuous if for every open set $U \subseteq \mathbb{R}$, the inverse $f^{-1}(U) = \{x \in \mathbb{R} \mid f(x) \in U\}$ is open in $\mathbb{R}$ (the open set definition).

Assume that f satisfies the $\epsilon - \delta$ definition of continuity. Then for $U \subseteq \mathbb{R}$, take any point $x \in f^{-1}(U) = \{x \in \mathbb{R} \mid f(x) \in U\}$. Since U is an open set, there must exist some $\epsilon > 0$ such that an open ball (B_ϵ) around $f(x)$ with radius ϵ is in U , that is, $B_\epsilon(f(x)) = \{y \in \mathbb{R} \mid |y - f(x)| < \epsilon\}$. So $B_\epsilon(f(x)) \subseteq U$.

Since f is continuous with the $\epsilon - \delta$ definition of continuity, $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$. Since distance is symmetric, $|f(x) - f(y)| < \epsilon$ is the same as $|f(y) - f(x)| < \epsilon$. Then, $f(y) \in B_\epsilon(f(x)) \subseteq U$, so $f(y) \in U$ and $y \in f^{-1}(U)$. Therefore, $|x - y| < \delta \Rightarrow y \in f^{-1}(U)$ and since every point in $f^{-1}(U)$ has some open ball around it contained in $f^{-1}(U)$, $f^{-1}(U)$ is open. And therefore, f is continuous in the open-set definition of continuity.

Question 6

Suppose $f : X \rightarrow Y$ is continuous. If x is a limit point of the subset A of X , is it necessarily true that $f(x)$ is a limit point of $f(A)$?

The statement is false. As a counterexample, choose $X = Y = \mathbb{R}$ with the standard topology. Let $f(x) = 6$ (as an example) for all $x \in X$. Then we take the subset $A = (6, 7) \subset \mathbb{R}$, $f(A) = \{6\}$ by our example with the limit point $x = 6$. Obviously f is continuous as it is a trivial function mapping $\mathbb{R}$ to 6.

However, $f(x = 6) = 6$ is not a limit point of $f(A = (6, 7)) = \{6\}$ because $f(A)$ contains only a single point.

Question 7

Let $F : X \times Y \rightarrow Z$. We say that F is continuous in each variable separately if for each y_0 in Y , the map $h : X \rightarrow Z$ defined by $h(x) = F(x \times y_0)$ is continuous, and for each x_0 in X , the map $k : Y \rightarrow Z$ defined by $k(y) = F(x_0 \times y)$ is continuous. Show that if F is continuous, then F is continuous in each variable separately.

Proof: Let $F : X \times Y \rightarrow Z$ be a continuous function. We must show that F is continuous in each variable separately.

Fix a point $y_0 \in Y$ and define a new function $h : X \rightarrow Z$ as $h(x) = F(x, y_0)$. Then consider the map $g : X \rightarrow X \times Y$ defined by $g(x) = (x, y_0)$. g must be continuous since it maps $x \mapsto x$ and y_0 was fixed. Then,

$$F \circ g = F(g(x)) = F((x, y_0)) = h(x)$$

Similarly, fix a point $x_0 \in X$ and define a new function $h' : Y \rightarrow Z$ as $h'(y) = F(x_0, y)$. Then consider the map $g' : Y \rightarrow X \times Y$ defined by $g'(y) = (x_0, y)$. g' must be continuous since it maps $y \mapsto y$ and x_0 was fixed. Then,

$$F \circ g' = F(g'(y)) = F((x_0, y)) = h'(y)$$

Since both F and g are continuous functions and h is the composition of those functions, h must also be continuous. Therefore, F must be continuous in each variable separately.

Question 8

Let $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be defined by the equation

$$F(x \times y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } x \times y \neq 0 \times 0 \\ 0 & \text{if } x \times y = 0 \times 0 \end{cases}$$

1. Show that F is continuous in each variable separately.
2. Compute the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = F(x \times x)$.
3. Show that F is not continuous (using techniques of this class).

1.

We fix y_0 at some point in $\mathbb{R}$. If $y_0 = 0$, then $F(x, 0) = 0$ for all x so F is continuous in this case. If $y_0 \neq 0$, then $F(x, y_0) = \frac{xy_0}{x^2 + y_0^2}$. Since $x^2 + y_0^2 > 0$ for all x, y_0 , the denominator is never zero, so F is also continuous in this case. We can show the same is true for fixing a point x_0 to show that F is still continuous here, so F must be continuous in each variable separately.

2.

When $x \neq 0$,

$$g(x) = F(x \times x) = \frac{x^2}{2x^2} = \frac{1}{2}$$

And $g(x) = 0$ when $x = 0$.

3.

Proof: Consider the closed set $\{\frac{1}{2}\}$. If F is continuous, then the preimage of F should also be closed. $F^{-1}(\{\frac{1}{2}\})$ contains points in the form (x, x) . The point $(0, 0)$ is a limit point of $F^{-1}(\{\frac{1}{2}\})$ because it intersects at a point other than itself. $F(0, 0) = 0$ is not in the preimage, however, so the preimage is not closed as it does not contain all its limit points. Therefore, F must not be continuous.

Question 9

Let $\mathbb{R}^\infty$ be the subset of $\mathbb{R}^\omega$ consisting of all sequences that are "eventually zero," that is, all sequences $(x_1, x_2, \dots)$ such that $x_i \neq 0$ for only finitely many of i . values of i . What is the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ in the box and product topologies? Justify your answer.

In the **box topology**, the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ is $\mathbb{R}^\infty$. We will show that $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$ by showing that its complement is open. Let $x \notin \mathbb{R}^\infty$, so there are infinitely many nonzero coordinates in x . Consider the neighborhood $U = \prod_i U_i$, where we choose each U_i by restricting $\mathbb{R}$ to be an open interval around x_i that excludes 0 whenever $x_i \neq 0$. So for some $\epsilon > 0$,

$$U_i = \begin{cases} (x_i - \epsilon, x_i + \epsilon) & x_i \neq 0 \\ \mathbb{R} & x_i = 0 \end{cases}$$

Then each $y \in U$ has infinitely many nonzero coordinates, so $y \notin \mathbb{R}^\infty$. Thus U is an open neighborhood of x disjoint from $\mathbb{R}^\infty$, so x is not a limit point. Therefore, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$ in the box topology.

In the **product topology**, the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ is $\mathbb{R}^\omega$. Let $x \in \mathbb{R}^\omega$. We will show that every open set containing x intersects $\mathbb{R}^\infty$. Consider the neighborhood $U = \prod_i U_i$, where $U_i = \mathbb{R}$ for all finitely many i . Then we define the sequence y by $y_i = x_i$ for all i smaller than some number N , and $y_i = 0$ for all i greater than N .

$y \in \mathbb{R}^\infty$ since the sequence is eventually 0. Also $y \in U$ since $y_i = x_i \in U_i$ for $i \leq N$ and $y_i = 0 \in \mathbb{R} = U_i$ for $i > N$. Thus every open neighborhood of x contains a point in $\mathbb{R}^\infty$, so x is in $\overline{\mathbb{R}^\infty}$. Therefore, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$ in the product topology.

Question 10

Given sequences $(a_1, a_2, \dots)$ and $(b_1, b_2, \dots)$ of real numbers with $a_i > 0$ for all i , define $h : \mathbb{R}^\omega \rightarrow \mathbb{R}^\omega$ by the equation

$$h((x_1, x_2, \dots)) = (a_1x_1 + b_1, a_2x_2 + b_2, \dots)$$

Show that if $\mathbb{R}^\omega$ is given the product topology, h is a homeomorphism of $\mathbb{R}^\omega$ with itself. What happens if $\mathbb{R}^\omega$ is given the box topology?

Proof: Let $\mathbb{R}^\omega$ have the product topology. Given sequences $(a_1, a_2, \dots)$ and $(b_1, b_2, \dots)$ of real numbers with $a_i > 0$ for all i , we define the function $h : \mathbb{R}^\omega \rightarrow \mathbb{R}^\omega$ by the equation

$$h((x_1, x_2, \dots)) = (a_1x_1 + b_1, a_2x_2 + b_2, \dots)$$

Then, each $h_i = a_ix_i + b_i$ for all i . Since h_i is a projection which is continuous in the product topology, and $a_ix_i + b_i$ itself is also continuous, h itself must be continuous. Furthermore, because $a_i > 0$, h is bijective and we can define the continuous inverse:

$$h^{-1}((y_1, y_2, \dots)) = \left(\frac{y_1 - b_1}{a_1}, \frac{y_2 - b_2}{a_2}, \dots \right)$$

Therefore, h is a homeomorphism of $\mathbb{R}^\omega$ to itself in the product topology.

Now let $\mathbb{R}^\omega$ have the box topology. Then, for $U = \prod_i U_i$

$$h(U) = h\left(\prod_i U_i\right) = \prod_i (a_iU_i + b_i)$$

Each $a_iU_i + b_i$ is open in $\mathbb{R}$, so $h(U)$ must also be open. We can also define the open inverse as

$$h^{-1}\left(\prod_i V_i\right) = \prod_i \left(\frac{V_i - b_i}{a_i}\right)$$

Therefore, h is a homeomorphism of $\mathbb{R}^\omega$ to itself in the box topology.